

Experiment No. 05

Aim: Verification of Kirchhoff's voltage law.

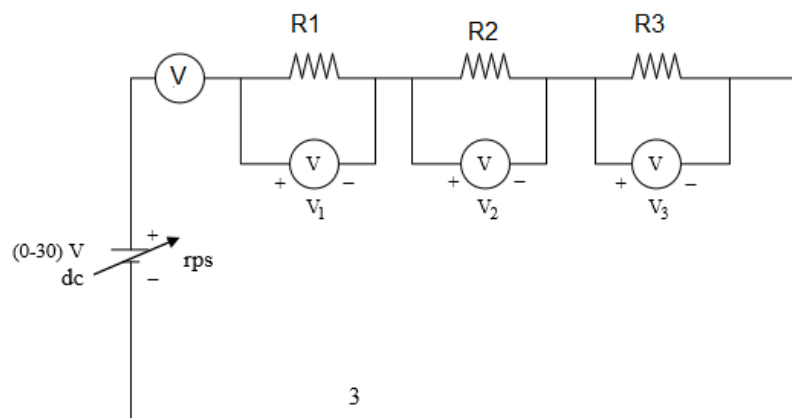
APPARATUS REQUIRED: 30V dc supply, Digital Multimeter as a voltmeter, Resistance Connecting Wires.

THEORY:

Kirchhoff's second rule (Kirchhoff's Voltage Law or KVL Loop rule): **It states that the algebraic sum of all potential drops and emfs along any closed path in a network is zero.** OR. The algebraic sum of the emfs in a loop of a circuit is equal to the algebraic sum of the product of current and resistances in it. Kirchhoff's voltage law is based on the assumption that there is no fluctuating magnetic field linking the closed loop. This is not a safe assumption for high-frequency (short-wavelength) AC circuits. In the presence of a changing magnetic field the electric field is not a conservative vector field. Therefore the electric field cannot be the gradient of any potential. That is to say, the line integral of the electric field around the loop is not zero, directly contradicting Kirchhoff's voltage law.

It is often possible to improve the applicability of Kirchhoff's voltage law by considering "parasitic inductances" (including mutual inductances) distributed along the conductors. These are treated as imaginary circuit elements that produce a voltage drop equal to the rate-of-change of the flux.

CIRCUIT DIAGRAM:



$$V = V_1 + V_2 + V_3$$

PROCEDURE:

1. Kirchhoff's Voltage law:
 1. Connect the circuit as shown in fig.
 2. Measure the voltages across the resistors.
 3. Observe that the algebraic sum of voltages in a closed loop is zero.

OBSERVATION TABLE:**KVL:**

S.NO	Voltage (V)	Theoretical			Practical			
		V1	V2	V3	V1	V2	V3	V=V1+V2+V3
1								
2								
3								

CALCULATION:

$$R \text{ total} = R1 + R2 + R3$$

$$I = V/R \text{ for}$$

$$V = \text{Voltage}$$

$$I =$$

$$V1 = I \cdot R1 = \quad V, \quad V2 = I \cdot R2 = \quad V, \quad V3 = I \cdot R3 = \quad V, \quad V = V1 + V2 + V3 = \quad V$$

RESULT: Hence we have verified KVL.

PRECAUTIONS:

1. Avoid loose connections.
2. Keep all the knobs in minimum position while switch on and off of the supply.
3. Input voltage increases minimum range.

VIVA QUESTIONS:

1. What is another name for KVL?
2. Define network and circuit?
3. What is the property of inductor and capacitor?